A Framework to Determine the Digital Skills Preparedness of Graduates for Industry

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***Abstract*—Rapid technological advancements are disrupting traditional job markets necessitating job seekers to develop new skillsets suitable for the digital economy. This phenomenon has a major impact on the economy, particularly in developing countries. Consequently, it is crucial for institutional offerings to be aligned to industry requirements in every discipline of higher education, in order to ultimately sustain and improve the economy. This paper presents a framework designed to determine the alignment of the digital skills that students acquire from higher education to the digital skills requirements of industry. This alignment will aid higher education institutions in improving the digital skills preparedness of their graduates, and ultimately sustaining the digital economy. Given that the digital economy requires its employees to possess a specific level of digital skills, which may vary in each sector, the proposed framework is therefore not discipline specific. Consequently, this framework may employed to establish an alignment between any discipline in higher education and the respective industry that its graduates feed into. The authors have systematically reviewed related articles to determine the factors influencing the digital skills preparedness of graduates for industry. Relevant studies were analyzed, thereby resulting in the development of a digital skills preparedness model.**

Keywords—Digital Skills, Preparedness Framework, Teaching & Learning Curricula, Institutional Challenges & Transformation

I. INTRODUCTION

Effective use of digital technology requires specific skills and talents. It is advocated that those who possess the skills to comprehend and interact with digital information to manipulate, arrange, and present it according to their understanding, possesses digital intelligence [1]. The concept of 'digital skill' has been defined in several disciplines, and as a result the terms, 'skills', 'literacy', 'competence', 'knowledge' and 'fluency' all refer to digital abilities [2]. Gekara, et al. [3] defines digital skills as "a combination of a digital mind-set (hardware, software, information, systems, security and innovation), knowledge (theoretical comprehension and understanding), competence (cognitive and practical knowhow) and attitude (value and beliefs)".

Ideally, higher education institutions prepare students with the necessary digital skills required by industry. Thereafter, these graduates find employment in the relevant sectors. Consequently, the unemployment rate drops, the country's digital economy is globally competitive and the country prospers, holistically. However, this can only be achieved if the ideologies and/or policies of government, higher education and industry are in alignment. The following authors concur that the actual circumstances are far from ideal, worldwide. Students work hard to gain a tertiary education, by taking loans and making sacrifices, in hope of attaining a job. Nevertheless, the shocking reality is that the unemployment rate of graduates in South Africa, for example, is at its highest of 453,330 (7.3%), since September 2003 [4]. After months of being unemployed, some graduates have resorted to begging for jobs at the street corners [5]. South Africa and other developing countries, face this major problems, as the growing unemployment rate negatively affects the economy, amongst other things.

In preparation for the working world, academic institutions should strive to improve their graduates' digital skills preparedness in all disciplines. Therefore, the aim of this study is to present a theoretical framework that aids in the alignment between the digital skills requirements of industry and the digital skills preparedness of graduates entering the workplace. This framework can be applied to any discipline in higher education and correlated to the respective sector within industry. For example, the digital skills requirements of organisations in the "business, real estate and finance sector" first needs to be understood. Thereafter, the digital skills of tertiary students entering this sector need to be evaluated, in order to determine the digital skills preparedness of those students. Furthermore, it is also important to explore the teaching and learning curricula of tertiary institutions regarding digital skills; the tertiary institutions policies and procedures on the digital skills of their students; as well as the challenges faced by tertiary institutions that deter institutions from efficiently and/or effectively producing digitally skilled graduates for industry.

The 'ripple effect' of employing this framework is that tertiary institutions will better understand the digital skills requirements of industry and align their curricula to meet those requirements. Consequently, graduates entering the workplace will possess the digital skillsets that is required for their sector. Additionally, these graduates will be better suited to current and future industry requirements, ultimately sustaining and improving the economy [6, 7] .

The next section discusses relevant literature that aided the development of the digital skills preparedness framework. The methodology section explains the procedure used to develop the proposed framework, and the results and discussions section provides a detailed explanation of the variables employed in the framework.

II. RELATED WORK

Information and Communications Technology (ICT) has become pervasive in contemporary societies as means of advancing educational systems, sustaining economic development via the invention of different products and services, affording access to information and proficiency to support developments in education, agriculture and health, as well as connecting communities, educators and learners [8]. In order to utilize ICT, one would not only need access to the

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Internet and the technology, but also requires the digital skills necessary to use the technology effectively [8]. However, the following research suggests that there is a gap between the digital skills preparedness of graduates entering the workplace and the digital skills requirements of the various industry sectors [6, 7, 9-11].

A. The Digital Skills Crisis

The education system continuously fails to produce a cohort of young individuals who are trained to apply technology in day-to-day activities and who are prepared for the workplace once they leave the education system [10]. As a result, there is an urgent unfulfilled need for digital skills, which is predicted to get worse [10]. Therefore, there is a need for substantial and continuous investment in training and education at all levels to reduce this digital skills gap. This means that higher education institutions need to play a greater role in improving the digital skills of their students. In order to reduce the digital skills gap, it is imperative that there is an alignment between the digital skills requirements of industry and the digital skills preparedness of graduates entering the job market.

Craffert, et al. [9] highlighted that South Africa is lagging behind regarding its digital skills readiness, in contrast to the revolutionizing paradigm of a progressively digitally infused society and an increasingly digitalized economy. Due to the current digital skills shortage in industry, Craffert, et al. [9] strongly recommends that institutions of higher education prepare students with digital skills to enable them to work proficiently and effectively, thereby ensuring that future employees are digitally skilled. Furthermore, training students in multi-literacies provides students with media, information and digital skills, which helps them to associate their academic experiences to those in real-life contexts [6, 11]. Hence, universities need to restructure their degree programs, in order to incorporate the core digital skills that will prepare students for the workplace.

Craffert, et al. [9] found that the likely role of industry to guide educational institutions is being under-utilized. Subsequently, it is strongly advocated that partnerships be made between educational institutions and industry in order to guarantee that the necessary skills are supplied to sustain a digital economy [9]. At the G20 Summit, Chetty, et al. [7] proposed that the G20 requires a general digital skills support strategy, which identifies the fundamental digital skills needed by employers in developing economies, like South Africa. They explain that digital training is a core component in people harnessing enhanced usage of investments in infrastructure and technology.

B. The Impact of Digital Skills on the Economy

ICT and digital literacy is deemed an essential competence for complete involvement in an information society and a knowledge economy [11]. A South African director of Google, emphasizes that the Internet is the focal point of economic growth, and more digital skills programs are needed, which should be aimed at helping the people of Africa contribute to the digital economy [12]. Sector Education and Training Authorities (SETA) have identified skills development as a crucial requirement for South Africa's economic growth [13]. In the present highly competitive knowledge economy, there exists exponential increases of information easily available via novel technologies that are continuously evolving at an astonishing rate [6].

To be 'literate' and prolific participants in the present knowledge economy, students should graduate with the 'functional and critical thinking skills' in Information, Media and Technology that will help them to obtain a satisfying career, as well as to contribute as productive citizens to the knowledge economy [6]. He emphasizes that graduating students who do not possess these skills will be significantly disadvantaged, in a time where there is global competition for these skills.

Most developed countries have already established, or are in the process of establishing a digital economic strategy [14]. World economists, all agree that a digitally skilled workforce is necessary in order to benefit from the opportunities presented by the digital economy [14]. The European Union has successfully standardized, assessed and certified digital skills with their 'International Computer Driving License' (ICDL) course [14]. Simply being in possession of technological infrastructure, such as hardware, software and telecommunication, is insufficient to offer a competitive advantage and to prosper in the digital economy [14]. However, it is crucial to possess a workforce that can control these vital assets effectively and efficiently.

C. Digital Skills in Teaching and Learning

Over the centuries, foundational paradigms have provided frameworks that directed teaching and learning, specifically related to the Industrial Age where specialization, memorization and compartmentalization augmented productivity and profits [15]. However, in the last two decades, the pervasive infiltration of technology into daily life has transformed the way people live, in addition to the approaches used by the 21st century occupations and industries, to function [15]. In order to remain competitive, each industrial firm needed to embrace technology or fall out of the profitability pursuit [15]. This phenomenon compelled firms to introduce novel tools into their manufacturing processes and require its new employees to possess new skills [15]. This is how the 21st century fundamental skills requirements emerged [15].

Vītoliņa [16] explored the methods to enable students to employ the digital skills acquired in practice and to assess the impact of students' prior knowledge level. Her findings indicate that the skills developed are used pragmatically if the knowledge level prior to learning a new topic is higher than the average level [16]. She suggests that prior training is required as individuals reach an acceptable knowledge level. Additionally, the effect of technology enhancement positively influences higher-order thinking skills, as measured by verbal intelligence [17]. Consequently, it is imperative that each discipline, specifically in higher education, align the digital skills preparedness of its students to the respective industry sector.

Even the healthcare industry is rapidly evolving in-line with the demand for improved flexibility in the educational approach in our rapidly developing society [18]. Healthcare is undoubtedly a fast-growing industry, and graduates who choose to move in healthcare, ought to ensure they possess sound digital literacies, especially as they relate to professional communication [18]. Therefore, the continuous advancement of technologies compels student nurses to

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acquaint themselves with digital skills and technologies to guarantee their use in a healthcare context is effective, safe and user-friendly [18]. It is thus vital that every discipline in in higher education align its curricular to its respective industry, as developing countries are facing the consequences of the digital skills gap.

D. Plans and Proposals for Digital Skills

Some sectors have a plan prepared to bridge the digital skills gap. For example, the skills plan for the banking sector has indicated the following implications for digital skills [10]. Firstly, there is a greater demand for advanced digital skills as a general driver of change. Since many business units are now integrated, staff need to be knowledgeable of technology across various business units and disciplines, it is therefore crucial to invest more on digital skill advancement initiatives. Additionally, new employees of the banking sector must be familiar with technology and the change in the way the banking sector now does business. Owing to technology advancements, scarce skills are no longer directly related to banking, as they now comprise process engineers, computer programmers and typically individuals in the ICT industry. Clearly, the banking sector mostly employs digital skills and these implications show the significant impact of those skills on the organisations within the sector [10].

Even as a developed country, the United Kingdom is determined to increase the digital skills of the workforce, as well as its general population [19]. For example, Higher Education includes national initiatives, such as the 'Developing Digital Literacies' programme, the 'Digital Literacies in the Disciplines' programme and the 'Changing the Learning Landscape programme' to name a few [20]. The key motivation for cultivating the digital skills of students have been acknowledged as a necessity by employers and professional bodies, as well as in Government strategies and policies [20, 21]. Chetty, et al. [7] strongly suggest that developing countries, like South Africa, require digital skills support strategies, which identifies the fundamental digital skills needed by employers in developing economies. Developing countries need to learn from the developed countries on how to bridge the digital skills gap, effectively.

III. METHODOLOGY

Numerous studies were systematically reviewed to determine the factors influencing the digital skills preparedness of graduates for industry. In total, 692 articles were assessed from which 23 articles closely met the inclusion criteria. Studies relating to digital skills of students in higher education were analyzed, thereby resulting in the development of a digital skills preparedness model (see Fig. 1).

Table 1 is the concept matrix of recent studies that influenced the development of the digital skills preparedness model. Some of the articles contributed to more than one variable. The variables identified in the concept matrix are considered core components with regard to the alignment of the digital skills required by industry to the digital skills preparedness of graduates.

TABLE 1. CONCEPT MATRIX

Variables	Article
Digital skills acquired through tertiary education	Chetty, et al. [22] Becker, et al. [23] Blackwood [24]
Teaching and Learning Curricula of the Institution	Kivunja [15] Vītoliņa [16] Becker, et al. [23] Funes, et al. [25]
Policies and Procedures for Tertiary Institutions	Kivunja [15] Tiatri and Jap [17] Evangelinous, et al. [20] Chetty, et al. [22] Becker, et al. [23] Davies, et al. [26] Johansson [27]
Challenges Faced by the Institution	Mosupi [12] South African Government [13] SAQA [28] Oyedemi [29] Becker, et al. [23]
Required Transformation of the Institution	Becker, et al. [23] Oliver [30] Davies, et al. [26] Ainley, et al. [8] Moloney and Farley [18]
Digital skills preparedness of students for industry	Becker, et al. [23] Alexander, et al. [31] Davies, et al. [26] Mills, et al. [32] Leckie, et al. [33] Calonge and Shah [34] Jensen, et al. [35] Brown [36]

IV. RESULTS AND DISCUSSION

A. The Proposed Digital Skills Preparedness Framework

In order to prepare students for industry demands, they need to be given access to curriculum that meets the demands of the 21st century by emphasizing the development of digital skills and competencies [37]. The institutional policies and procedures also need to be studied to ensure that they are aligned to the demands of industry. Institutional challenges ought to have a negative impact on the digital skills preparedness of students; therefore, it is essential to understand these challenges, so that appropriate measures can be taken. The institutional transformation of a university is important to improve its digital skills offerings to better suit industry requirements, and ought to have a positive impact on the digital skills preparedness of students.

Chetty, et al. [22] advocates that is imperative to recognise the correlation between the 'demand' for digital skills requirements by industry and the 'supply' of digital skills by higher education institution's training programmes. They concur that tertiary institutions must be responsive to adapt to the digital skills requirements of industry. Even though this alignment between industry and higher education is complex, it is essential for developing countries to make every effort to adapt [22]. Their study focuses more on educating learners in the school system to be adequately prepared for the workplace. However, thousands of graduates spend millions of Rand each year in developing countries, like South Africa, but are still unemployed after years of sacrifice and hard work. Consequently, more emphasis needs to be placed on finding resolutions to break this cycle soon, as people are beginning to lose faith in our education system, especially at tertiary level [22].

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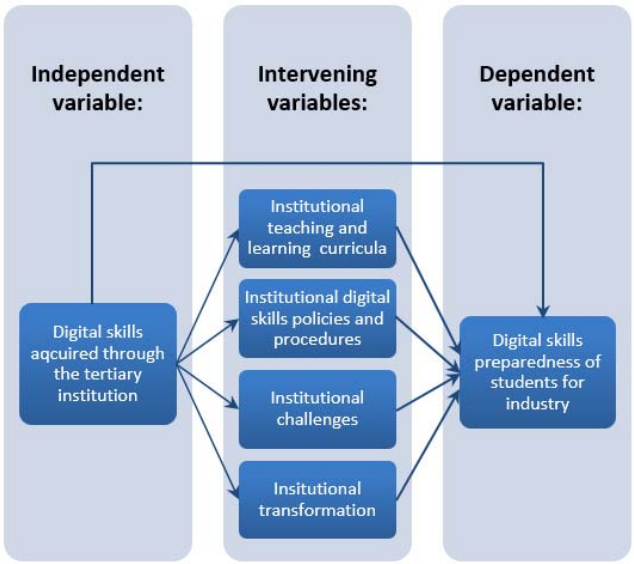


Fig. 1. Digital Skills Preparedness Model

Since each sector may require a different set of digital skills, Fig. 1 can be applied to any industry sector of any country to determine the digital skills preparedness of students entering a particular sector.

1) Independent Variable

The digital skills acquired through tertiary institutions is the independent variable of Fig. 1, as it will be manipulated to determine its impact on the dependent variable, which is the digital skills preparedness of students for industry. The dynamic digital skills requirements obtained from industry will enable higher education institutions to align their teaching and learning curricula to meet industry requirements [22]. Once these requirements are incorporated in higher education training programs, the graduates will most likely obtain formal employment in his/her field of study. This will produce a ripple effect, whereby employment opportunities would be far greater, thereby creating greater incentives for students to register for tertiary education and maybe even higher degrees [22].

2) Intervening Variables

The digital skills preparedness of students for industry is affected by a few intervening variables shown in Fig. 1, which will influence the relationship between the digital skills acquired through the tertiary institution and the digital skills preparedness of students for industry. The significance of each intervening variable in relation to the digital skills preparedness of graduates for industry, is discussed below.

a) Teaching and Learning Curricula of the Institution

The teaching and learning curricula of an institution (i.e. the first intervening variable in Fig. 1) plays a pivotal role in the digital skills preparedness of students for industry. The growing demand for the labour force to be digitally competent coerces higher education institutions to evaluate their curricula and guarantee that digital skills are embedded as a graduate trait [20, 21]. In order to produce students equipped with digital skills to help them become proficient in the 21st century globalized trades, professions and occupations, it is crucial that digital technologies are embedded in pedagogy and curriculum [15, 20]. After studying the learning paradigms that have directed pedagogy over time, Kivunja [15] asserts that a shift in curriculum and pedagogy is needed, towards a paradigm that highlights problem solving and critical thinking. Typically, digital skills are developed by teaching and learning, therefore, it is imperative to offer such hands-on training on digital skills, particularly if individuals gain from acquiring new skills [16]. For learning of digital skills, a variety of suitable learning resources must be provided, which the learner can choose to read or watch several times, subject to the student's needs and experience [16]. Brown [36] maintains that digital skills are necessary for survival, production and creativity, embracing digital and open resources, learning through virtual and physical environments, and research in real-time.

b) Policies and Procedures for Tertiary Institutions

Policies and procedures adopted by an institution (i.e. the second intervening variable in Fig. 1) should aim to align the teaching and learning curricula to the digital skills requirements of industry. Establishing institutional quality-control policies and procedures will guarantee that students graduate with the essential digital skills required in their particular field [20]. Tiatri and Jap [17] maintains that the effect of technology enhancement positively influences higher-order thinking skills, as measured by verbal intelligence. The development of digital skills relative to their discipline offers the potential of inspiring graduates to become digital leaders and innovators within their disciplines [20]. Furthermore, measuring digital literacy and regularly guaranteeing that policies and procedures are adequately agile to respond to dynamic nature of digital skills' will lead to efficiency across a country [22]. In order to provide students with an effective education in the 21st century, a pedagogical paradigm shift is necessary so that students are prepared with the skills required in the 21st century workplace, and must be also proficient in problem solving, thinking for themselves, working in teams and leading others to triumph in the Digital/Knowledge Economy [15].

c) Challenges Faced by the Institution

Some challenges faced by an institution (i.e. the third intervening variable in Fig. 1) ought to impede the digital skills preparedness of students, therefore it is considered an intervening variable. It is imperative that these institutional challenges are identified, so that the South African Department of Higher Education can present viable solutions. The South African Qualifications Authority (SAQA) have not catered for the development of digital skills in their South African National Qualifications Framework [28]. South Africa acquiring the technology and infrastructure from developed countries does not imply that the users will possess the necessary digital skills to make effective use of the investments in the technology and infrastructure by the government and industry [12]. Furthermore, existing cross-national studies of digital skills have largely been undertaken in developed countries with greater levels of ICT resources and ICT development [13].

Consequently, as a developing country, South Africa needs to determine if university students possess the core digital skills required by industry [12]. University students are predominantly relevant to the study of digital skills, since most South African universities provide students with access to computer laboratories, and studies show that education is a major indicator of digital skills [29]. It is thus necessary to understand the challenges faced by tertiary

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institutions, so that correctional measures can be developed/taken to ensure that the teaching and learning curriculum, as well as policies and procedures cater for the digital skills upliftment of tertiary students to ultimately produce a digitally skills workforce.

d) Required Transformation of the Institution

The required institutional transformation (i.e. the fourth intervening variable in Fig. 1) is essential in order to better align the institution's teaching and learning to meet the digital skills requirements of industry. In order to understand the degree and nature of institutional transformation required, it is important to first establish the following, respectively:

- the teaching and learning curricula on digital skills;
- the policies and procedures of tertiary institutions on digital skills;
- the challenges faced by tertiary institutions that deter them from efficiently and/or effectively producing digitally skilled graduates.

The transformation of the institution is necessary if its digital skills alignment with industry is poor or non-existent. The institutional transformation will deal with strategies to: (1) improve the teaching and learning curricula; (2) update the policies and procedures; and eliminate the challenges face by the institution.

3) Dependent Variable

Essentially, the degree of digital skills acquired through tertiary institutions will explain the variance in the digital skills preparedness of students for industry. The European Commission advocated that 90% of jobs may soon require basic digital skills in accountancy, medicine, architecture, engineering, art, nursing and many more careers [19, 38]. The UK, as a first world country, has identified the impact of the digital revolution on the country and has created reports/policies to help its citizens improve their digital skills. This in turn will benefit UK's economy, holistically. Research suggests that skill-biased technological change is rapidly transferred from the developed countries to the developing countries [39]. Therefore, South Africa, as a developing country, needs to follow suit or else our economy will continue to suffer.

The difference between the dependent and independent variable in Fig. 1 is that the independent variable is the digital skills that students acquire through tertiary institutions. However, the dependent variable is the digital skills knowledge that is required at industry level. From a primitive perspective the digital skills acquired by students at tertiary institutions in South Africa are designed to have a strong alignment to industry expectations (as illustrated in the direct relationship between the independent and dependent variables in Fig 1). However, a more informed examination of this relationship reveals a significant contribution by a set of intervening variables that also has an influence on the relationship between the independent and dependent variables (also illustrated in Fig.1).

V. Conclusion

Technology advancements necessitates graduates entering the workplace to possess the necessary digital skills. Thus, it is vital that digital skills development of tertiary students, as well as infrastructure development be given the same priority, if these graduates entering the industry sectors are to compete in the digital economy. Furthermore, international educators, government, organizations and institutions all see the need to develop and implement plans to improve the digital skills of students and the workforce, with the aim of sustaining the economy. In response to the issues discussed, the proposed digital skills framework will assist researchers in bridging the gap between industry requirements and academia curricula concerning digital skills. The framework may be applied to each discipline in order to align its teaching and learning curricula, as well as its institutional policies and procedures to the digital skills requirements of its respective industry.

VI. Recommendations

Further research may be conducted to test the proposed framework using different methodologies. Additionally, this framework may be used to investigate the digitals skills requirements in various other disciplines in order to investigate the digital skills preparedness of students entering other industry sectors. Once tested, the digital skills preparedness model may be used as a theoretical framework as the basis of future studies.

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